

Part A - Manual Computation

Given Image (5x5)

$$I = \begin{bmatrix} 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \\ 10 & 10 & 10 & 80 & 80 \end{bmatrix}$$

Sobel - x Kernel

$$K = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}$$

1. Output Feature Map size

$$\text{Output size} = \frac{N - F}{S} + 1$$

$$N = 5 \text{ (Image size)}$$

$$F = 3 \text{ (Kernel size)}$$

$$S = 1 \text{ (Stride)}$$

$$\text{Output size} = \frac{5 - 3}{1} + 1 = 3$$

$$\text{Output size} = 3 \times 3$$

2. First output location (1,1)

Kernel placed at

$$I_1 = \begin{bmatrix} 10 & 10 & 10 \\ 10 & 10 & 10 \\ 10 & 10 & 10 \end{bmatrix}$$

Multiplying ~~each~~ element by element $I \otimes K$

$$\begin{aligned} & (10 \times -1) + (10 \times 0) + (10 \times 1) \\ & + (10 \times -2) + (10 \times 0) + (10 \times 2) \\ & + (10 \times -1) + (10 \times 0) + (10 \times 1) \end{aligned}$$

$$= -10 + 0 + 10 - 20 + 0 + 20 - 10 + 0 + 10 \\ = 0$$

$$\therefore \text{Output}(1,1) = 0$$

3. Complete output Matrix

Position (1,2)

$$I_2 = \begin{bmatrix} 10 & 10 & 80 \\ 10 & 10 & 80 \\ 10 & 10 & 80 \end{bmatrix}$$

Calculation:

$$(10 \times -1) + (10 \times 0) + (80 \times 1) \\ + (10 \times -2) + (10 \times 0) + (80 \times 2) \\ + (10 \times -1) + (10 \times 0) + (80 \times 1)$$

$$= -10 + 80 - 20 + 160 - 10 + 80$$

$$= 280$$

Position (1,3)

$$I_3 = \begin{bmatrix} 10 & 80 & 80 \\ 10 & 80 & 80 \\ 10 & 80 & 80 \end{bmatrix}$$

Calculation:

$$-10 + 0 + 80 \\ -20 + 0 + 160 \\ -10 + 0 + 80 \\ = 280$$

Since image rows are identical

Final output matrix

$$\begin{bmatrix} 0 & 280 & 280 \\ 0 & 280 & 280 \\ 0 & 280 & 280 \end{bmatrix}$$

A. Analysis

The strong sobel response - 280
appear b/w : 10 $\rightarrow$ 80

↓
vertical edge

Left side of image has constant intensity:

$\therefore$ Sobel respon ≈ 0

$$\begin{bmatrix} 08 & 01 & 01 & 10 & 10 \\ 08 & 01 & 01 & 01 & 01 \\ 08 & 01 & 01 & 01 & 01 \end{bmatrix}$$

Hence

Edge location : b/w 3rd and 4th column

Flat regions : value close to zero because
nearby pixels have nearly
same intensity.

$$\begin{bmatrix} 08 & 01 & 01 \\ 08 & 01 & 01 \\ 08 & 01 & 01 \end{bmatrix}$$

$$\begin{bmatrix} 08 & 01 & 01 \\ 08 & 01 & 01 \\ 08 & 01 & 01 \end{bmatrix}$$